

The Original Balanced Mellin Kernel

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1 The original balanced Mellin kernel: full polynomial comparison and meromorphic remainder

The authoritative shared transcript constructs the raw Mellin realization in A1167, retains its kernel complex in A1361, and replaces it by the $\mathbb{C}[t]$ -balanced realization in A1427. This section computes the polynomial action on the remaining kernel on those original objects. It uses no replacement arithmetic function and no auxiliary zero packet.

1.1 Original objects and full-plane Euler inverses

Retain, exactly as in A1167 and A1427,

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}, \text{ (BK.1)} \\ g(s) &= 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \\ H_\phi(s) &= \frac{2\pi^{s/2}M_+\phi(s)}{s(s-1)\Gamma(s/2)}. \end{aligned}$$

The original Fourier convention is $\widehat{\phi}(\nu) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \nu} dx$. The original theta construction proves

$$\mathcal{M}\Theta\phi = gH_\phi, \quad \Theta D = D\Theta, \quad \widehat{D\phi} = (1-D)\widehat{\phi}, \quad H_{D\phi}(s) = sH_\phi(s), \quad H_{\widehat{\phi}}(s) = H_\phi(1-s). \quad \text{(BK.2)}$$

The last equality follows directly from Poisson's identity $\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x)$, Mellin substitution, and $g(1-s) = g(s)$; cancellation holds on the nonempty open set $g \neq 0$ and then throughout the plane by holomorphy. Here H_ϕ is entire: Taylor subtraction of the even smooth function at zero gives only simple possible poles of $M_+\phi$ at the negative even integers; the corresponding zeros of $1/\Gamma(s/2)$ cancel them. At $s = 0$, $M_+\phi$ is holomorphic because $\phi(0) = 0$, and $s\Gamma(s/2)$ tends to 2. At $s = 1$, $M_+\phi(1) = 0$ is exactly the retained integral condition. In particular

$$H_\phi(1) = 2(M_+\phi)'(1). \quad \text{(BK.3)}$$

The topology and all these maps are the original Schwartz ones.

For every $\rho \in \mathbb{C}$ one has the exact range formulas

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)V = \ker(H_{(\cdot)}(\rho)). \quad \text{(BK.4)}$$

Both operators are injective. We now prove the full-plane source formula, including the endpoint $\rho = 1$ that cannot be handled by dividing by $1 - \rho$.

On the target zero-Mellin domain the inverse is the original integral

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy. \quad (\text{BK.5})$$

Differentiating gives $(D - \rho)S_\rho F = F$. Since the total Mellin moment is zero, the same expression is $-x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ near zero. For $\sigma = \Re \rho$, an upper bound $|F(y)| \leq C_N y^{-N}$ yields $|S_\rho F(x)| \leq C_N x^{-N}/(N - \sigma)$ when $N > \sigma$. A lower bound $|F(y)| \leq C'_N y^N$ yields $|S_\rho F(x)| \leq C'_N x^N/(N + \sigma)$ when $N + \sigma > 0$. The differential identity controls each further Euler derivative. Thus every original seminorm is finite and the inverse is continuous on the indicated kernel. The only homogeneous solution is $cx^{-\rho}$, which belongs to $\mathcal{B}$ only when $c = 0$. Integration by parts proves $\mathcal{M}((D - \rho)F) = (s - \rho)\mathcal{M}F$, so the range is exactly the first kernel in (BK.4).

For the source, first let $\Re \rho > 0$ and $H_\phi(\rho) = 0$. For $\rho \neq 1$, the multiplier of $M_+\phi(\rho)$ in (BK.1) is finite and nonzero, so this Mellin moment vanishes. For $\rho = 1$ the same moment already vanishes by the definition of V . The integral (BK.5) is consequently also

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du. \quad (\text{BK.6})$$

For every ordinary derivative of order k , differentiating this integral inserts $u^k \phi^{(k)}(xu)$. Uniform domination on bounded x intervals follows from $\Re \rho + k > 0$. Therefore the result extends smoothly and evenly across zero with value zero there. The upper-tail expression gives Schwartz decay, with every derivative, at infinity. If $\rho \neq 1$, integration of $(D - \rho)S_\rho \phi = \phi$ over $\mathbb{R}$ gives

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi = \int_{\mathbb{R}} \phi = 0.$$

If $\rho = 1$, integration by parts in a right half-plane gives

$$M_+(S_1 \phi)(s) = \frac{M_+\phi(s)}{s - 1}.$$

Taking its removable value at 1 and using (BK.3) gives $\int_{\mathbb{R}} S_1 \phi = 2(M_+\phi)'(1) = H_\phi(1) = 0$. Thus the original source integral condition is retained at every ρ with positive real part.

If $\Re \rho \leq 0$, put $\lambda = 1 - \rho$, so $\Re \lambda \geq 1$. Fourier preserves V , and (BK.2) gives $H_{\widehat{\phi}}(\lambda) = H_\phi(\rho) = 0$. The exact inverse is

$$S_\rho^V \phi = -\widehat{S_\lambda \phi}. \quad (\text{BK.7})$$

On even functions Fourier squares to the identity. The intertwiner in (BK.2) therefore gives $(D - \rho)(-\widehat{S_\lambda \phi}) = \phi$, with every source condition preserved. For positive real part the inverse is (BK.6). The homogeneous solution cannot be Schwartz at infinity unless zero, so the inverse is unique. Identity (BK.2) proves the reverse containment in the second formula of (BK.4). This completes the full-plane calculation without excluding either endpoint or negative even Taylor coordinate.

1.2 Every polynomial kernel and cokernel is computed on the actual theta sequence

Set $A = \mathbb{C}[t]$, with t acting as the original D , and $Q = \mathcal{B}/\Theta V$. For a nonconstant monic polynomial

$$h(t) = \prod_{\rho} (t - \rho)^{n_{\rho}}, \quad E_h = A/(h),$$

let j_h retain all Taylor jets of orders less than n_{ρ} . Iterating (BK.4) gives

$$h(D)V = \ker(j_h H), \quad h(D)\mathcal{B} = \ker(j_h \mathcal{M}). \quad (\text{BK.8})$$

Indeed after dividing by $D - \rho$, the Mellin multiplier or H is divided by $s - \rho$ by (BK.2); the remaining zero orders are exactly the orders needed at the next division. Injectivity proves that the iterated inverse is independent of the chosen ordering.

The source jet map is onto because the original vector

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \quad \text{satisfies} \quad H_{P(D)\phi_*} = P. \quad (\text{BK.9})$$

The target jet map is onto because its functionals on compactly supported smooth logarithmic functions are the independent functions $y^j e^{\rho y}/j!$. For precision, a dependence vanishing on an interval extends to an entire exponential polynomial. Applying $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{n_\sigma}$ removes the other summands; its restriction to $e^{\rho y}$ times polynomials of degree less than n_ρ is invertible, since it is a product of nonzero constants plus nilpotent derivatives. Then $(\partial_y - \rho)$ repeatedly separates the polynomial coefficients. All coefficients vanish. If the finite vector-valued jet map were not onto, a nonzero linear functional on its missing quotient would give such a dependence, a contradiction.

Consequently the quotient square and exact theta sequence are

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{M_{j_h g}} & E_h, \end{array} \quad (\text{BK.10})$$

$$0 \longrightarrow Q[h(D)] \xrightarrow{\delta_h} E_h \xrightarrow{M_{j_h g}} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{BK.11})$$

These identities now hold for every polynomial h , without a strip restriction. The map δ_h sends $[F]$ to $j_h H_\phi$ where $h(D)F = \Theta\phi$. The source ϕ is unique because Θ is injective. Changing F by a theta relation changes ϕ by an $h(D)$ multiple, so the map is well-defined. If its value is zero, (BK.8) writes $\phi = h(D)\psi$ and injectivity of $h(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. A vector in the kernel of $M_{j_h g}$ is represented by a polynomial P ; the zero-jet condition permits applying the actual inverse of $h(D)$ to $\Theta(P(D)\phi_*)$, producing its preimage. Quotienting the right target in (BK.10) proves the last cokernel. This proves the entire sequence with its original maps.

1.3 The balanced kernel has invertible polynomial action

Fix any analytic stalk $O = \mathcal{O}_{\mathbb{C}, s_0}$, with t acting on O by multiplication by s . Write

$$M = O \otimes_A Q, \quad N = O/(g), \quad b : M \rightarrow N, \quad b(a \otimes [F]) = [a\mathcal{M}F], \quad K = \ker b. \quad (\text{BK.12})$$

The map is balanced by (BK.2). The ring O is torsion-free over the PID A : a nonzero polynomial times a holomorphic germ is zero only when that germ is zero. It is therefore flat. One direct justification of this familiar PID fact is to write a torsion-free module as the directed union of its finitely generated submodules; each such submodule is free by the PID structure theorem. Tensoring with a free module preserves injections, and filtered colimits preserve exactness, proving flatness.

The map b is onto. In fact the original test function

$$F_0(x) = \exp[-(\log x)^2], \quad H_0(s) = \mathcal{M}F_0(s) = \sqrt{\pi} \exp(s^2/4) \quad (\text{BK.13})$$

belongs to $\mathcal{B}$ and H_0 is an invertible germ at every point. Euler derivatives of F_0 are polynomials in $\log x$ times F_0 , and $x^b F_0$ is bounded for every integer b . Its Mellin integral becomes $\int_{\mathbb{R}} e^{-y^2 + sy} dy$; real completion of the square gives the formula for real s , and entire continuation gives it for every complex s . Thus $b(aH_0^{-1} \otimes [F_0]) = [a]$, proving surjectivity.

For every nonzero polynomial h , the two induced maps are isomorphisms:

$$b : M[h] \xrightarrow{\sim} N[h], \quad \bar{b} : M/hM \xrightarrow{\sim} N/hN. \quad (\text{BK.14})$$

For a nonzero constant both statements are immediate. For a nonconstant polynomial flatness gives $M[h] = O \otimes_A Q[h(D)]$ and $M/hM = O \otimes_A (Q/h(D)Q)$. Also $O \otimes_A E_h = O/(h)$ by the explicit map $a \otimes [P(t)] \mapsto [aP(s)]$, with inverse $[a] \mapsto a \otimes 1$. The cokernel assertion in (BK.11) consequently gives

$$M/hM \cong O/(h, g) = N/hN,$$

and the isomorphism is precisely the Mellin map $\bar{b}$.

For the kernel map retain the actual local units. Put $z = s - s_0$ and write

$$h(s) = a(z)z^n, \quad g(s) = u(z)z^m, \quad a(0)u(0) \neq 0, \quad n, m \geq 0. \quad (\text{BK.15})$$

Under (BK.11), a local vector v in $\ker(g : O/(h) \rightarrow O/(h))$ corresponds to a torsion class with $hMF = gH_\phi$, and its image under b is exactly

$$v \mapsto \left[\frac{g}{h} v \right] \in N[h]. \quad (\text{BK.16})$$

This is well-defined: changing v by h times a germ changes the displayed product by g times a germ. The kernel condition makes the product holomorphic. Explicitly its domain is $z^{\max(n-m, 0)} O/z^n O$ and its target is $z^{\max(m-n, 0)} O/z^m O$. Formula (BK.16) is multiplication by the actual unit u/a , together with the power z^{m-n} . It sends precisely the indicated source powers to the indicated target powers and has inverse multiplication by a/u and the power z^{n-m} . It is an isomorphism, including $m = 0$ or $n = 0$ when both spaces are zero. This proves the first map in (BK.14) without dropping a local factor of g or a repeated-zero order.

Apply the kernel-cokernel exact sequence for multiplication by h to $0 \rightarrow K \rightarrow M \xrightarrow{b} N \rightarrow 0$. It is

$$0 \rightarrow K[h] \rightarrow M[h] \rightarrow N[h] \rightarrow K/hK \rightarrow M/hM \rightarrow N/hN \rightarrow 0. \quad (\text{BK.17})$$

Both maps in (BK.14) are isomorphisms, so $K[h] = 0$ and $K/hK = 0$. We have proved

$$\boxed{h : K \xrightarrow{\sim} K \quad \text{for every nonzero } h \in \mathbb{C}[t].} \quad (\text{BK.18})$$

This is a computed property of the original balanced kernel.

It has a stronger local form. Every nonzero holomorphic germ is a unit times z^r . Units already act bijectively, and (BK.18) applied to $(t - s_0)^r$ makes z^r bijective. Hence K has a unique module structure over the meromorphic germ field $\text{Frac } O$, obtained by taking these actual inverses:

$$\frac{a}{b} \cdot k = a(b|_K)^{-1}k. \quad (\text{BK.19})$$

If $a/b = c/d$, the equality $ad = bc$ and invertibility of bd show that the two prescribed actions agree. The field laws follow by multiplying through their nonzero denominators.

1.4 Canonical finite spectral part and an explicit nonzero kernel class

At a point where $m = \text{ord}_{s_0} g > 0$, (BK.14) for $h = (t - s_0)^m$ gives a unique inverse $j : N \xrightarrow{\sim} M[z^m]$ of b on that kernel. Therefore

$$M = M[z^m] \oplus K, \quad x = j(bx) + (x - j(bx)), \quad M[z^m] \cong O/(g). \quad (\text{BK.20})$$

The first summand is all the O -torsion of M . Indeed every torsion element is killed by a power of z . For larger powers than m , (BK.14) still gives the same unique preimage of each element of N , because K has no z -torsion. When $m = 0$, $N = 0$ and the formula is $M = K$. All the full nilpotent jets of an actual zero occur in the first summand of (BK.20).

The kernel K is nonzero. The only classical zero-set input needed for the following assertion is the already-retained Hardy theorem: there are distinct $T_j \rightarrow +\infty$ with $g(1/2 + iT_j) = 0$. The cumulative reader uses this exact input in `theta_gamma_reference.tex`, (TG.26), and

sum_connection_stieltjes.tex, (FC.21), with Hardy's original 1914 source. No assumption of RH is used.

Let $q_0 = [F_0] \in Q$. It is A -nontorsion. If $P(D)F_0 = \Theta\phi$ with a polynomial P , then (BK.2) and (BK.13) give

$$P(s)\sqrt{\pi}e^{s^2/4} = g(s)H_\phi(s).$$

At every $1/2 + iT_j$, the nonzero exponential forces $P(1/2 + iT_j) = 0$. A nonzero polynomial cannot have infinitely many distinct roots, so $P = 0$. Thus $A \rightarrow Q$, $P \mapsto P(D)q_0$, is injective. Flatness now proves that $O \rightarrow M$, $a \mapsto a \otimes q_0$, is injective. In particular

$$0 \neq g \otimes q_0 \in K. \quad (\text{BK.21})$$

The corresponding cycle is explicit on the original balanced complex. Put $b_* = \Theta\phi_*$, so $\mathcal{M}b_* = g$. In $O \otimes_A \mathcal{B}$, set

$$Z_0 = g \otimes F_0 - H_0 \otimes b_*. \quad \beta Z_0 = gH_0 - H_0g = 0. \quad (\text{BK.22})$$

Its degree-one class is exactly the nonzero class in (BK.21), since its second term is the original theta boundary. For the original invariant source label $W_* = \mathbb{C}[D]\phi_*$, the map $O \otimes_A W_* \rightarrow O$ induced by H is an isomorphism by (BK.9). Thus the kernel complex at that label has degree-zero term zero, and (BK.22) is an explicitly nonzero degree-one kernel cycle. Passing to the full label V preserves its nonzero class, as proved by the A -nontorsion argument. This resolves the status left open in A1427: balancing removes the earlier displayed spurious relation, but its full degree-one kernel is not zero.

This statement does not create a spectral obstruction to RH. For every finite-dimensional A -module E , let p_E be the characteristic polynomial of its t action. Cayley–Hamilton gives $p_E E = 0$. If $u : E \rightarrow K$ is A -linear, then $p_E u(E) = 0$ and injectivity in (BK.18) gives $u = 0$. If $v : K \rightarrow E$ is A -linear, every $k \in K$ is $p_E k'$ for some k' , and $v(k) = p_E v(k') = 0$. Consequently

$$\text{Hom}_A(E, K) = 0 = \text{Hom}_A(K, E) \quad (\dim_{\mathbb{C}} E < \infty). \quad (\text{BK.23})$$

The entire kernel has no finite-dimensional invariant subspace and no nonzero finite-dimensional equivariant quotient. This does not assert the same for arbitrary subquotients of its proper submodules. Its calculated contribution to every full finite spectral jet observation is zero, whereas (BK.20) proves that the actual finite spectral torsion survives bijectively.

Finally every construction occurs in the original supported quotient. The structure assertions (BK.18)–(BK.23) concern the full source label V used in (BK.12). They are not asserted for an arbitrary proper D -invariant source label. For such a label $W \subset V$, the exact comparison that must be retained is

$$0 \longrightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]} Q_W \longrightarrow Q \longrightarrow 0, \quad Q_W = \mathcal{B}/\Theta W. \quad (\text{BK.24})$$

Injectivity of Θ proves the first map injective, and the quotient definitions prove its image is the last map's kernel. All maps commute with D when W is invariant. Tensoring this sequence with the flat O retains the entire kernel $O \otimes_A (V/W)$, which can have additional finite spectral torsion before transport to the full label. No such proper-label torsion is discarded by the scope of the structure theorem. The inclusion of labels $W_* \hookrightarrow V$ carries $\mathcal{B}/\Theta W_* \rightarrow \mathcal{B}/\Theta V$ with the original kernel $\Theta V/\Theta W_*$. Balancing imposes the explicit linear relations $sa \otimes v - a \otimes Dv$ within each such label. A linear map L at a fixed label ℓ lifts as $(\ell, x) \mapsto (\ell, Lx)$ and sends external τ to external τ . Addition joins labels and applies their given linear transports; the displayed formulas commute with those transports. Therefore (BK.22) is sent to the supported zero $(\ell, 0)$ by Mellin evaluation, while its nonzero kernel class remains at that label. Equations (BK.18)–(BK.24), including every actual nilpotent jet and the proper-label transition kernel, describe precisely which full-source kernel assertions survive the complete original quotient diagram.